

GOVERNMENT ENGINEERING COLLEGE, RAJKOT

MID-SEM EXAMINATION (B.E.)

SEM-III (EC, CE, CIVIL, AI&DS)

SUBJECT: Probability and Statistics

Total Marks: 30

DATE: 15/09/2025

SUBJECT CODE: BE03000251

Time: 11.30 AM TO 12.45 PM

Que 1: Do as directed. [Attempt any two]

[10]

- (1) There are three boxes of identical appearance, each containing 4 spark plugs. It is known that the box 1 contains only one defective spark plug, box 2 contains two defective spark plugs while all the four spark plugs of the box 3 are non defective. A spark plug is drawn at random from a box, selected at random, is found to be non defective. What is the probability that it came from the box 1? (CO-1) [5]
- (2) Consider a class with 20 students. 10 students passed in Maths, 15 in English, & 13 in both. What is the probability that a student passed in (i) either Maths or English, (ii) neither Maths nor English? (CO-1) [5]
- (3) For the given data, $P(A/B)=2/3$, $P(B)=1/5$ & $P(A \cup B)=1/2$. Find (i) $P(A \cap B)$ & (ii) $P(A)$. (CO-1) [5]
- (4) Let X be the random variable of getting number of heads when three coins are tossed simultaneously. Find the mean and the variance for the random variable X. (CO-1) [5]

Que 2: Do as directed. [Attempt any two]

[10]

- (1) The mean and the variance of the Binomial distribution are 3 and 1.2 respectively. Find n, p and $P(X \geq 1)$. (CO-2) [5]
- (2) A manufacturer knows from experience that the resistance of resistors he produces is normal with a mean $\mu = 100$ ohms and standard deviation $\sigma = 2$ ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms? ($P(0 \leq Z \leq 1) = 0.3413$) (CO-2) [5]

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- (3) The average time it takes to serve a customer at a petrol pump is 6 minutes. The service time follows exponential distribution. Calculate the probability that A customer will take (i) less than 2 minutes to complete the service and (ii) between 4 and 5 minutes to complete the service. (CO-2) [5]
- (4) If a random variable has a Poisson distribution such that $P(X=1)=P(X=2)$, find (i) the mean of the distribution and (ii) $P(X \geq 1)$. (CO-2) [5]

Que 3: Do as directed. [Attempt any two]

[10]

(1) Fit a straight line $y = ax + b$ for the following data.

x	0	1	2	3	4
y	1.0	2.9	4.8	6.7	8.6

(CO-5) [5]

(2) Fit a parabola $y = ax^2 + bx + c$ to the following data.

x	10	12	15	23	20
y	14	17	23	25	21

(CO-5) [5]

(3) Fit a curve $y = a e^{bx}$ to the following data by using method of least squares.

x	1	5	7	9	12
y	10	15	12	15	21

(CO-5) [5]

(4) Fit a curve $y = a b^x$ to the following data.

x	2.5	3.2	5.7	8.1	11.6	14.2	15.8
y	0.373	0.358	0.310	0.275	0.226	0.193	0.179

(CO-5) [5]